

Multi-Objective Optimisation

Summer Term 2026

Exercise Sheet – Case Study

General information

The case study will be graded on a pass/fail system. **You need to pass; otherwise you won't be admitted to the final exam!** Exceptional submissions will be granted 3 extra points towards the final exam, but the exam grade without these extra points should be passing. You should work in groups of three to five students; submissions with fewer or greater number of authors won't be accepted!

Upload your submission (a single ZIP-archive including a single PDF of your compiled case study report and your documented Python code) using the upload area in the PANDA room. Only a single student per group needs to submit.

The hard deadline is Friday, July 10th, at 6pm Paderborn time!

Project description

The goal of this case study is to implement and benchmark the multi-objective evolutionary algorithms that we have discussed in the lectures and the tutorials, such as NSGA-II (Deb et al. 2002) or SMS-EMOA (Beume, Naujoks, and Emmerich 2007).

You should implement at least two algorithms in python by yourself. **Hint:** use the code from the exercises as a starting point. Then, choose at least 3 different benchmark problems from the benchmark problem sets DTLZ or WFG, which are implemented in the Python library `pymoo`, and evaluate the performance of your implementation with the decision dimension of $n = 10$. Evaluate the performance on using at least two different settings for the number of objectives (for example, $p = 2$ and $p = 3$).

Recall what we learned about randomised search heuristics: due to random choices on many stages of the optimisation process you need to run each algorithm multiple times on each problem with meaningful parameters.

Use the dominated Hypervolume and at least one other indicator to compare the algorithm performance (`pymoo` implements quite a handful of indicators). In addition, report on the runtime of each algorithm across different objective counts. Present and visualize the results in a meaningful way (e.g., show box-plots of the distribution of indicators split by problem and algorithm) and interpret your observations. Use your knowledge of the strengths and weaknesses of your implemented algorithms to try to explain the difference in performance on the different test problems.

We suggest to consider the paper *A Tutorial on the Performance Assessment of Stochastic Multiobjective Optimizers* by Knowles et al. (Knowles, Thiele, and Zitzler 2006) provided in PANDA for more information on performance assessment.

Compile the results into a report given the MALEO- $\LaTeX$ report template provided in the PANDA course. The page limit is 8 pages and the minimum size for the report is 6 pages. This includes references, figures and tables. The template itself provides a quick guide to $\LaTeX$.

Your report should be self-containing and consist out of the following sections:

1. Title and Authors, where you should make sure to include the names, student numbers, and email addresses of all members of your group.
2. Introduction, where you briefly introduce the problem you are solving and how you plan to do so.
3. Background, where you should briefly describe the problems and algorithms you are benchmarking. Feel free to describe the strengths and weakness of each algorithm, and how you believe they will perform on your chosen problems.
4. Methodology, where you should describe the setup of your benchmarking environment. This description should be detailed enough that someone who reads your report would be able to reproduce your work.
5. Results, where you show the results of your benchmarking. In addition, you should describe why you think a certain algorithm performed well or poorly, even if it is just an educated guess.
6. Conclusion, where you briefly sum up your work.

An Abstract is not necessary for the report.

References

- Deb, K. et al. (2002). "A fast and elitist multiobjective genetic algorithm: NSGA-II". In: *IEEE Transactions on Evolutionary Computation* 6.2, pp. 182–197. DOI: [10.1109/4235.996017](https://doi.org/10.1109/4235.996017).
- Beume, Nicola, Boris Naujoks, and Michael Emmerich (2007). "SMS-EMOA: Multiobjective selection based on dominated hypervolume". In: *European Journal of Operational Research* 181.3, pp. 1653–1669. DOI: [10.1016/j.ejor.2006.08.008](https://doi.org/10.1016/j.ejor.2006.08.008).
- Knowles, J., L. Thiele, and E. Zitzler (Feb. 2006). *A Tutorial on the Performance Assessment of Stochastic Multiobjective Optimizers*. 214. revised version. Computer Engineering and Networks Laboratory (TIK), ETH Zurich, Switzerland. DOI: [10.3929/ethz-b-000023822](https://doi.org/10.3929/ethz-b-000023822).